

- **Bash Scripting Topics: Beginner to Functions:**

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✓1. Introduction to Bash Scripting

What is Bash? Bash is a Unix shell and command language used for scripting and automating tasks in Linux.

How to write a shell script:

- Create a new file, e.g., `script.sh`
- Add commands to it

How to make it executable:

```
chmod +x script.sh
```

Shebang:

```
#!/bin/bash
```

Example:

```
#!/bin/bash  
echo "Hello, World!"
```

✓2. Comments

- Single-line comment: `#`
- Multi-line comment :: `<<'COMMENT' ... COMMENT`

Example:

```
# This is a comment  
echo "Running Script..."
```

✓3. Variables

- Defining and using variables
- Rules: no spaces around `=` and variable names must be alphanumeric or underscores

Example:

```
name="Abhishek"  
echo "Hello, $name"
```

✓4. User Input

- Use read to take input from user

Example:

```
read -p "Enter your name: " username
echo "Welcome, $username"
```

✓5. Conditional Statements

- if statement

```
num=10
if [ $num -gt 5 ]; then
    echo "$num is greater than 5"
fi
```

- if-else statement

```
num=3
if [ $num -gt 5 ]; then
    echo "$num is greater than 5"
else
    echo "$num is less than or equal to 5"
fi
```

- if-elif-else statement

```
num=5
if [ $num -gt 5 ]; then
    echo "$num is greater than 5"
elif [ $num -eq 5 ]; then
    echo "$num is equal to 5"
else
    echo "$num is less than 5"
fi
```

✓6. Comparison Operators

- Integer: -eq, -ne, -lt, -le, -gt, -ge
- Strings: =, !=, -z, -n

Example:

```
str1="hello"
str2="world"
if [ "$str1" != "$str2" ]; then
    echo "Strings are different"
fi
```

✓7. Loops

For Loop

```
for i in 1 2 3; do
    echo "Number $i"
done
```

While Loop

```
count=1
while [ $count -le 3 ]; do
    echo "Count $count"
    ((count++))
done
```

Until Loop

```
num=1
until [ $num -gt 3 ]; do
    echo "Number: $num"
    ((num++))
done
```

✓8. Case Statements

Example:

```
read -p "Enter a number (1-3): " num
case $num in
    1) echo "You chose One";;
    2) echo "You chose Two";;
    3) echo "You chose Three";;
    *) echo "Invalid";;
esac
```

✓9. Arrays

Example:

```
fruits=("apple" "banana" "cherry")
echo "First fruit: ${fruits[0]}"
```

✓10. Functions in Bash

- Creating and calling functions

Example:

```
greet() {  
    echo "Hello, $1!"  
}
```

```
greet "Abhishek"
```

Return values using ** or **:

```
add() {  
    result=$(( $1 + $2 ))  
    echo $result  
}
```

```
sum=$(add 3 5)  
echo "Sum is: $sum"
```
